Lesson Plan · Monday, November 9

Algebra I · Mr. Pandey · Module 3, Topic 1, Lesson 2: Literal Equations (Day 1 of 2) · essentials only (✪)

Objectives (students will…)

- Find the slope, x-intercept, and y-intercept from slope-intercept, point-slope, and standard form.
- Pick the most efficient form for each feature.
- Rewrite linear equations from one form to another.

TEKS

- **A.3A** slope from an equation in $y = mx + b$, $Ax + By = C$, $y - y_1 = m(x - x_1)$
- A.2B write linear equations in different forms
- A.12E literal equations
- Process: A.1D

Key terms

- slope-intercept form · point-slope form · standard form

Materials

- Slides: m3t1-lesson2-literal-equations (.html/.pptx) · publisher slides with answers: BLMathematics_AlgI_M3_T1_L2_LessonSlides_ENG_with_answers.pptx
- Bell work: BELL WORK/T1-Bell-Work-Mon-11-09.pdf + key (online: m3t1-lesson2-bellwork-day1)
- Exit ticket: EXIT TICKET/T1-Exit-Ticket-Mon-11-09.pdf + key (written like the End of Topic 1 test)
- Video: videos/M3T1-Lesson2-Literal-Equations.mp4 · IEP and 504 versions of all of these: IEP/, Section 504/, and the -IEP / -504 PDFs
- Student Edition pp. 357–361 (Getting Started, Activity 2.1)
- Lesson 1 homework (SE pp. 355–356) to check
- Anchor chart: Key Characteristics of Linear Functions (slope, x-int, y-int)
- Whiteboards; exit ticket half sheets

Bell work (on the board)

- What value of x makes $2(x + 3) = x - 4$ true? A. -10
- How many solutions does $3x + 1 = 3x - 5$ have? G. No solution
- What are the slope and y-intercept of $y = -2x + 7$? C. slope -2, $(0, 7)$
- What is the x-intercept of $y = 2x - 6$? J. $(3, 0)$

Agenda (60 minutes)

Time	Activity	What to do / say
0–5	Bell work	Q1–2 review Lesson 1; Q3–4 preview today.
5–10	Homework check	Lesson 1 Assignment Practice 2: a none, b $x = 7$, d infinite, f infinite. Collect questions for later.
10–25	✪ Getting Started: Perimeter Perspectives · SE p. 358	Read the park scenario. Pairs write each perimeter equation: Juan $w = P/2 - \ell$, Mia $\ell = P/2 - w$. Q2: multiply each by 2 to get back to $2\ell + 2w = P$. **Say:** same formula, solved for the variable each person needs.
25–50	✪ Activity 2.1: Equations in Different Forms · SE pp. 359–361	Q1 $y = -\frac{2}{3}x + 4$: slope $-\frac{2}{3}$, $(0, 4)$, $(6, 0)$. Q2 $y = 4(x - 2)$: slope 4, $(0, -8)$, $(2, 0)$. Q3 $3x - 2y = 8$: slope $3/2$, $(0, -4)$, $(8/3, 0)$. Explicit teach the three forms (Q4). Q5 $Ax + By = C$: slope $-A/B$, x-int $(C/A, 0)$, y-int $(0, C/B)$. Q6: slope → slope-intercept or point-slope; a point → point-slope; y-int → slope-intercept. Pairs Q8: $y - 2 = 5(x - 7)$ $y = 5x - 33$; $8x - 2y = 12$ $y = 4x - 6$. Q9: $y + 7 = 0.5(x - 1)$ $x - 2y = 15$; $y = -4x + 3$ $4x + y = 3$. Q10 if time. **Test moves** (publisher slides 27–28, like test Q7, Q10): $4x + 3y = 12$ $y = -(4/3)x + 4$ (divide *every* term); $y + 4 = 3(x - 2)$ $y = 3x - 10$.
50–58	Exit ticket	1. Rewrite the equation in slope-intercept form. $6x + 3y = 12$ $y = -2x + 4$ 2. Rewrite the equation $y - 4 = 3(x + 2)$ in slope-intercept form. $y = 3x + 10$ 3. What are the x- and y-intercepts of $2x - 5y = 10$? $(5, 0)$ and $(0, -2)$ IEP/504: read aloud on request, properties/notes card, extra time.
58–60	Wrap up	Homework: Lesson 2 Assignment Practice 1a–d (SE p. 369). Tomorrow: literal equations (formulas).

Check for understanding

- Whiteboards: "Which form shows the y-intercept fastest?" slope-intercept
- Thumbs: "Standard form can have A = ½." Down: A, B, C are integers
- Activity 2.1 Q4–6 are the TE's check points.

After class

- Missed exit Q1 → Skills Practice Set A (standard → slope-intercept).
- Missed exit Q2 → Skills Practice Set B (point-slope →).
- Both correct → Skills Practice Set C.

All of Day 1 is starred. TE pacing is 2 days, and the district calendar matches (11/9–11/10).